

COVID-19 IMPACT, TESTING & VACCINATION ANALYSIS IN INDIA

DATA ANALYTICS PROJECT DOCUMENTATION

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Project Type:	Data Analytics Project
Dataset Source:	Kaggle

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1. Project Title

COVID-19 Impact, Testing & Vaccination Analysis in India

2. Project Objective

The main objective is to analyze the spread and severity of COVID-19 across Indian states over time. The project combines case, recovery, death, testing and vaccination indicators to identify major pandemic waves, high-burden states, testing patterns and vaccination progress. Python is used for data integration, cleaning, validation, EDA and visualization, while Power BI is used for interactive reporting.

3. Problem Statement

- COVID-19 impact varied considerably across Indian states and over different periods. Total cases alone could not explain the complete burden.
- Decision-makers needed to understand when major waves occurred, which states experienced the highest case and death burden, and where active cases were greatest.
- Testing patterns changed over time, making it important to understand how testing activity and positivity evolved.
- Vaccination progress needed to be examined alongside cases, deaths and testing to provide a broader view of the pandemic response.

Analytical Question: Which states and time periods experienced the greatest COVID-19 burden, how did testing and vaccination evolve, and what regions or periods should receive greater preparedness attention?

4. Dataset Information

Item	Project Detail
Dataset type	Integrated state-level COVID-19 analytical dataset
Original records	18,109
Records after cleaning	18,046
Final columns	23 (19 analytical + 4 documentation/status columns)
Geographical regions retained	36
Study period	30-Jan-2020 to 11-Aug-2021
Testing period	01-Apr-2020 to 10-Aug-2021
Vaccination period	16-Jan-2021 onward
Data Source	Kaggle

4.1 Important Columns

- Date, State
- Confirmed, New_Cases, Active_Cases
- Cured, New_Recovered
- Deaths, New_Deaths
- TotalSamples, Positive, Negative, Positivity_Rate
- Recovery_Rate, Case_Fatality_Rate
- Total_Doses, First_Dose, Second_Dose, Total_Individuals_Vaccinated

5. Tools and Technologies Used

Tool / Technology	Purpose
Python	Integration, cleaning, validation, outlier analysis, EDA and visualization
Pandas / NumPy	Data manipulation and validation
Matplotlib / Seaborn	Exploratory visualization
Microsoft Power BI	Interactive dashboard and reporting

6. Data Cleaning

1. Standardized column names, data types, and dates

Column names were standardized and the data was converted into appropriate data types. Date format was standardized for consistency.

2. Standardized state names

Inconsistent state names were corrected so the same state was represented uniformly.

3. Removed non-geographical administrative categories

Categories such as “Cases being reassigned to states” and “Unassigned” were removed because they do not represent actual geographical regions.

4. Removed duplicate records

Duplicate rows were checked and Date + State was validated so each state had only one record for a particular date.

5. Handled missing testing values carefully

Missing testing values were not treated as zero because missing means information was unavailable, whereas zero would incorrectly indicate no testing.

6. Data Cleaning — Continued

6. Retained structural vaccination missingness

Missing vaccination values before the vaccination program started were retained because vaccination data was not applicable during that period. Missing values after vaccination began were documented for further analysis.

7. Handled first-observation nulls

For New_Cases, New_Recovered, and New_Deaths, only null values occurring at the first observation of a state were replaced with 0.

8. Retained negative daily values

Negative values in daily cases, recoveries, or deaths were retained because they can occur due to reporting corrections or revisions.

9. Handled logically impossible Active_Cases values

Negative Active_Cases values were changed to missing values rather than replacing them with zero and creating fabricated data.

10. Performed IQR outlier analysis

The IQR method was used to identify potential outliers in New_Cases, New_Recovered, New_Deaths, Active_Cases, and TotalSamples.

11. Retained genuine pandemic extremes

Identified outliers were not automatically deleted because extremely high values can represent genuine COVID-19 surges.

Final Data Quality

- 18,046 records
- 23 columns
- 36 geographical regions
- 0 full duplicate rows
- 0 duplicate Date + State combinations

Overall, the cleaning process focused on improving data consistency and quality without removing genuine pandemic information.

7. Data Analysis / Exploratory Data Analysis

The EDA focuses on COVID trend and wave analysis and state-wise impact and severity. The analysis examines major waves, peak daily cases, peak deaths, peak active burden, and the states with the highest total cases and deaths.

EDA Analysis Area	Key Questions
EDA 1 — COVID Trend & Wave Analysis	Major waves, peak daily cases, peak deaths and peak active burden
EDA 2 — State-wise Impact & Severity	States with highest cases, deaths, active burden and daily surges

7. Data Analysis / Exploratory Data Analysis — Key Analyses

1. Peak Daily COVID-19 Cases: First Wave vs Second Wave

Compares the highest number of daily new COVID-19 cases recorded during the First Wave and Second Wave in India to identify which wave had the higher peak.

2. COVID-19 Trend and Wave Analysis in India

Studies how COVID-19 cases changed over time to identify major waves, increases, decreases, and peak periods.

3. Daily New COVID-19 Cases in India

Shows daily new COVID-19 cases and helps identify fluctuations and the highest case-reporting dates.

4. Relative Severity of COVID-19 Waves

Compares waves using confirmed cases, deaths, and active cases to determine which wave had the greater overall impact.

5. Top 10 States by Total COVID-19 Deaths

Identifies the states with the highest total reported deaths and helps compare fatality burden.

6. Top 10 States by Total Confirmed COVID-19 Cases

Identifies the states with the highest total confirmed cases and compares the spread across India.

8. Key Insights and Findings

#	Insight	Interpretation
1	Two major waves	Two prominent COVID-19 waves occurred during the study period.
2	Sharper second wave	Wave 2 peaked at 414,188 daily cases, 323.1% above Wave 1.
3	Higher mortality burden	Peak daily deaths rose from 2,003 to 6,148.
4	Greater active burden	Peak active cases rose from about 1.02M to 3.75M.
5	Maharashtra led overall burden	Maharashtra ranked highest across key state-level burden comparisons.
6	Outliers need context	Extreme values were investigated and retained when they could represent genuine surges or reporting revisions.

9. Power BI Dashboard

To bring the analysis together, an interactive Power BI dashboard was created as a concise three-page reporting solution. The dashboard combines overall COVID burden, state-level severity, daily trends, testing activity and vaccination progress. Users can interact with date and state filters to explore the indicators from different perspectives.

Dashboard Pages: State Impact & Severity • India COVID-19 Overview • Testing & Vaccination

9.1 Dashboard Page 1 — State Impact & Severity

This page focuses on the geographical distribution and severity of COVID-19 across Indian states. It highlights the top states by deaths and confirmed cases, shows state locations on a map, and compares cases against deaths. The KPI cards summarize total confirmed cases, total deaths and peak active cases, making it useful for quickly identifying high-burden states.

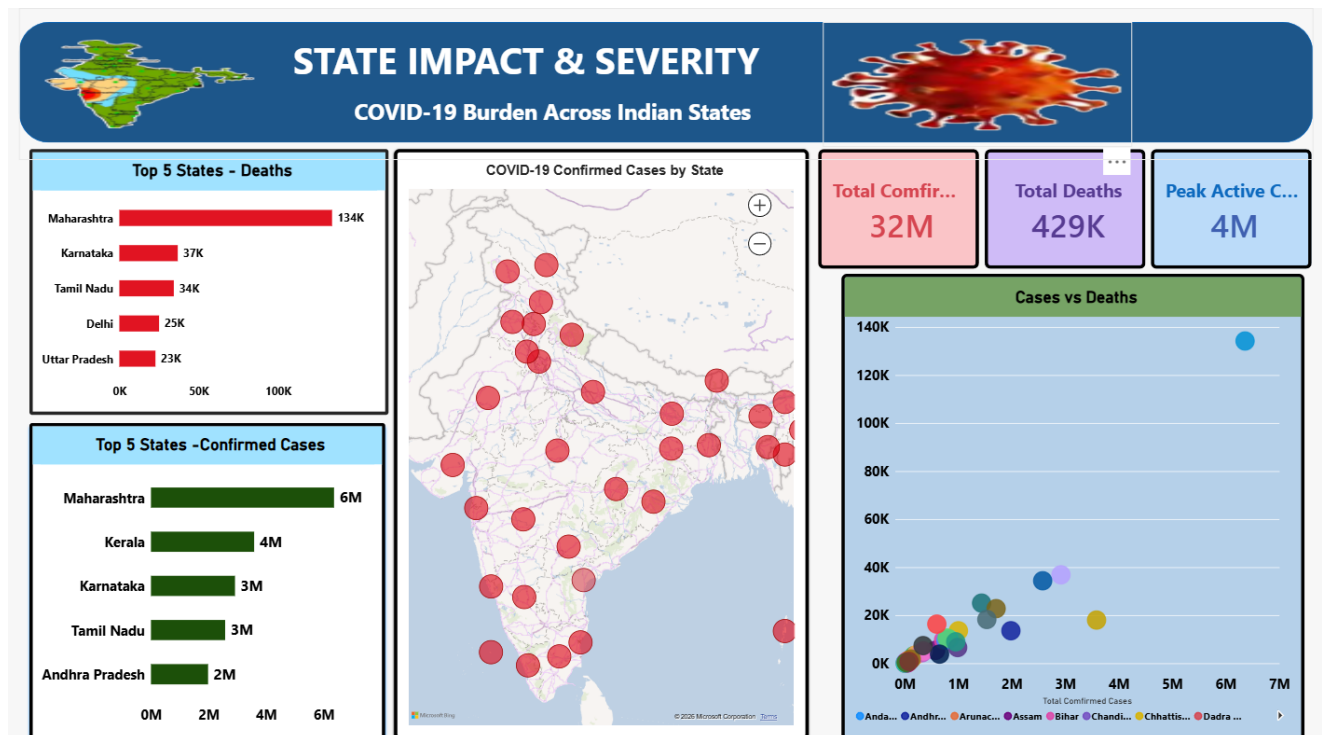


Figure 1: State Impact & Severity — COVID-19 Burden Across Indian States

9.2 Dashboard Page 2 — India COVID-19 Overview

The India COVID-19 Overview page provides a time-based view of the pandemic. The KPI cards summarize total confirmed cases, total recovered, total deaths and peak active cases. The daily new cases chart shows the major waves and peak reporting period, while the daily deaths chart highlights mortality changes over time. The active-cases trend shows how the number of active cases increased during major surges and subsequently declined.

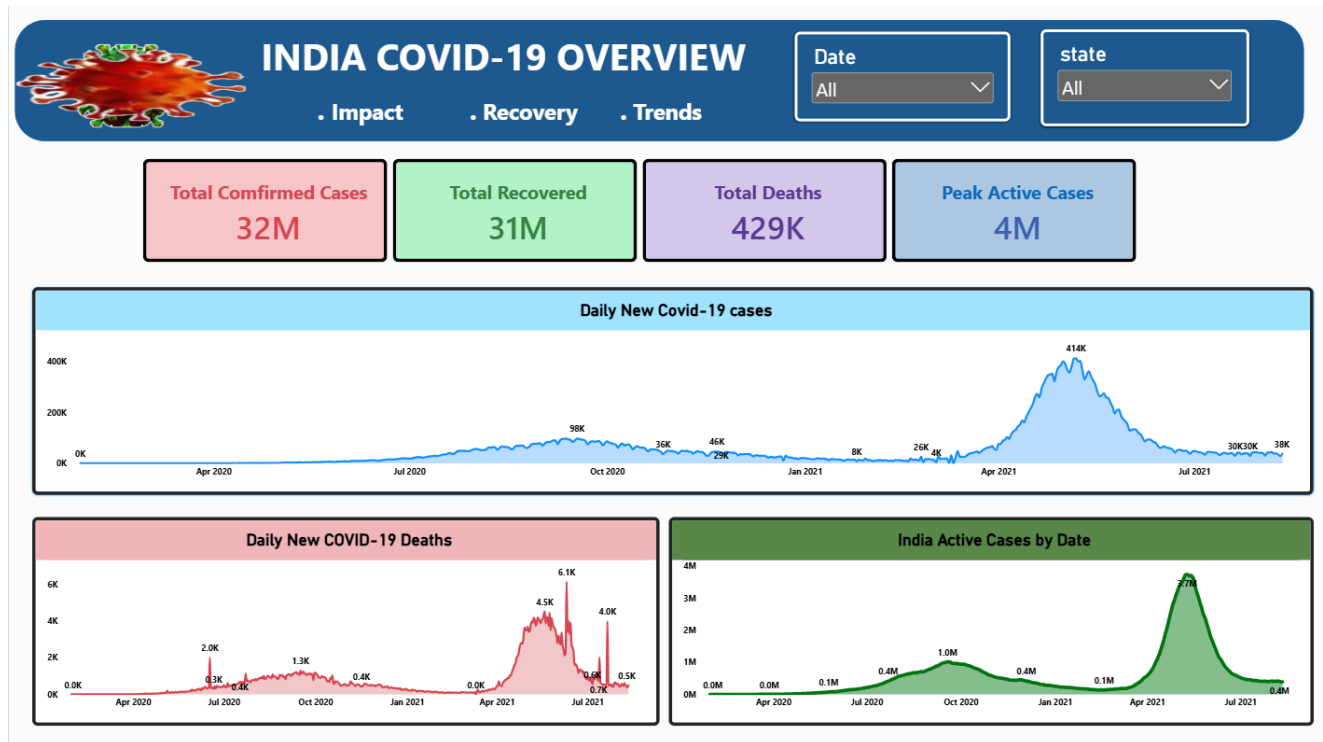


Figure 2: India COVID-19 Overview — Impact, Recovery and Trends

9.3 Dashboard Page 3 — Testing & Vaccination

The Testing & Vaccination page brings together testing intensity, positivity and vaccination progress. The KPI cards show latest total samples, latest positive tests, latest total doses and the overall positivity rate. The vaccination chart tracks cumulative vaccination progress, while the first-dose versus second-dose donut chart shows the distribution of doses. The positivity-rate trend helps interpret changes in test positivity, and the testing trend shows how cumulative testing expanded through the study period.

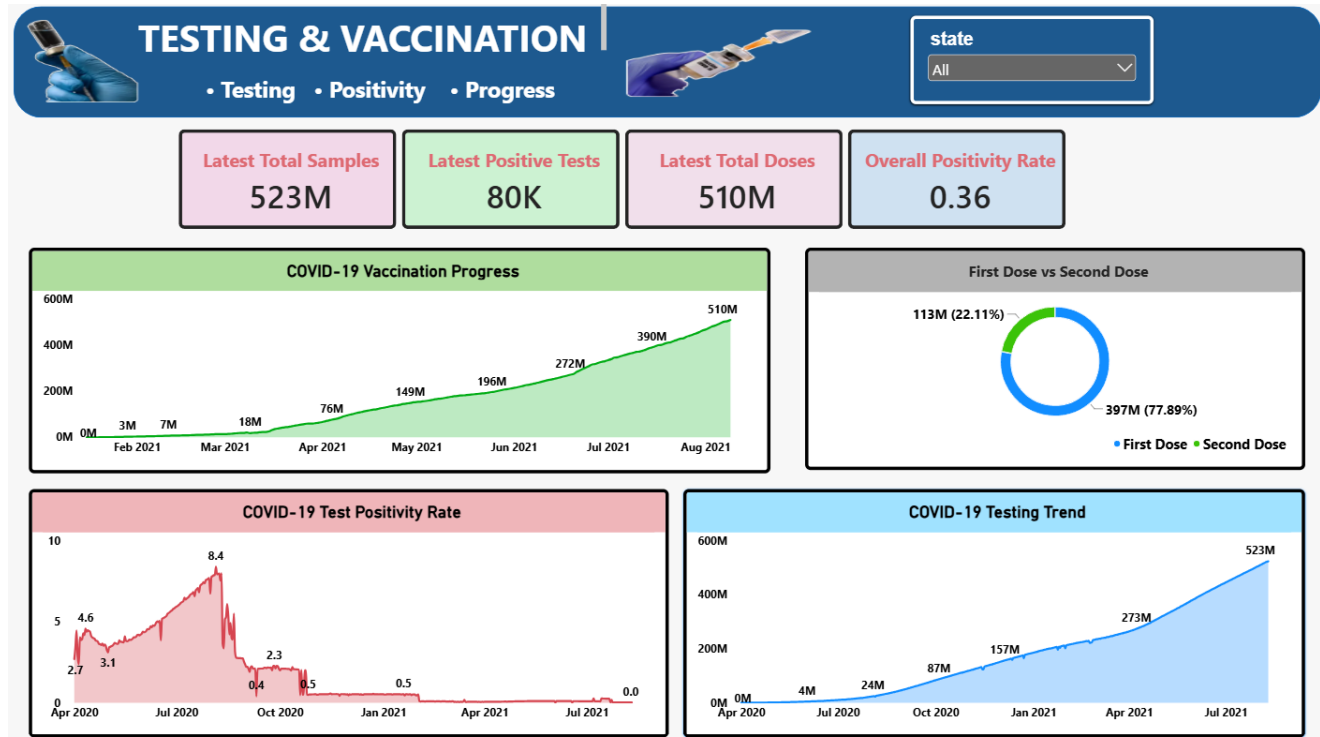


Figure 3: Testing & Vaccination — Testing, Positivity and Progress

Dashboard Purpose

Together, the three dashboard pages transform the cleaned analytical dataset into an easy-to-explore reporting view. They allow users to move from national trends to state-level severity and then to testing and vaccination progress, supporting clearer interpretation of pandemic burden and response patterns.

10. Conclusion

This project demonstrates how Python and Power BI can transform multi-source COVID-19 data into a structured analytical solution. Python was used to integrate, clean, validate and explore the data, while Power BI presents the results through a concise three-page interactive dashboard. The analysis identified two major waves, with the second wave producing substantially higher peaks in daily cases, daily deaths and active cases. State-level analysis showed Maharashtra carrying the highest overall burden across several major indicators. The project also demonstrates that missing values and extreme observations must be interpreted in context rather than automatically treated as errors.

11. Future Improvements

- Extend the dataset with later pandemic periods and booster information.
- Add population-normalized indicators such as cases/deaths/vaccinations per 100,000 population.
- Add district-level analysis where consistent data is available.
- Add healthcare-capacity indicators such as hospital beds, ICU availability and oxygen demand.
- Introduce forecasting or time-series modeling as an advanced analytics phase.
- Automate data refresh and dashboard updates.
- Investigate reporting revisions and data-quality anomalies using source metadata.

12. Project Developed By

Details	Information
Project Title	COVID-19 Impact, Testing & Vaccination Analysis in India
Project Type	Data Analytics Project
Team Name	Data Visionaries
Team Members	A.Naresh, J.Poojitha, J.Prasanna
Course / Program	Data Science
Institution	TEKS Academy
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Microsoft Power BI

13. References and Dataset Source

13.1 Dataset Source

The COVID-19 dataset used in this project was sourced from Kaggle. The dataset contains state-level information related to COVID-19 cases, recoveries, deaths, testing, and vaccination in India.

13.2 Project Resources

- COVID-19 Dataset – Used as the primary source for case, death, recovery, testing, and vaccination analysis.
- Python / Jupyter Notebook – Used for data integration, cleaning, validation, exploratory data analysis, and visualization.
- Power BI – Used to create the interactive dashboard and present the analytical findings.

13.3 Technologies Referenced

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Microsoft Power BI